

A new micropterous species of *Haematoloecha* from Vietnam (Hemiptera: Heteroptera: Reduviidae)

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Abstract

A new species of *Haematoloecha* Stål, 1874 (Hemiptera: Heteroptera: Reduviidae: Ectrichodiinae), *H. merkli* **sp. nov.**, is described from northern Vietnam based on micropterous males and females. A checklist of species placed into *Haematoloecha* and currently considered valid is presented.

Key words: taxonomy, new species, Palearctic Region, Indomalaya

Introduction

The ectrichodiine assassin bug genus *Haematoloecha* Stål, 1874 (Hemiptera: Heteroptera: Reduviidae) is distributed in East and Southeast Asia, from western China (Sichuan) throughout southern China and northern parts of Indochina (northern Vietnam, northern Laos) to the Korean Peninsula, Japan and Taiwan in the east. Currently, 18 species are considered valid (Maldonado Capriles 1990, Rédei & Tsai 2012).

Most species of the genus are known based on macropterous individuals only. However, two species were described based on micropterous individuals: *H. puetzi* (Kerzhner & Günther, 2004) (China: Sichuan) and *H. curta* Rédei & Tsai, 2012 (Taiwan) (Rédei & Tsai 2012). The present paper provides the description of an additional new species from northern Vietnam based on micropterous males and females.

Material and methods

Observations of all specimens were made using a Leica M205 C stereoscopic microscope. A Nikon D90 camera equipped with an AF-S Micro Nikkor 60 mm f/2.8G ED macro lens was used to take Figs. 1–6; the same camera with Nikon E Plan 4×/0.10 and Nikon Plan 10×/0.25 lenses set on a Nikkor 200 mm f/4 AI teleobjective was used for digital micrographs of higher magnification (Figs. 7–13). Male genitalia were dissected after short heating in hypertonic KOH solution. Drawings were made by using a camera lucida; measurements were taken using a calibrated ocular micrometer. All measurements are given in mm.

Taxonomy

Genus *Haematoloecha* Stål, 1874

Haematoloecha Stål, 1874: 48, 54. Type species by monotypy: *Scadra nigrorufa* Stål, 1867.

The taxonomic history of the genus was surveyed and a revised diagnosis was provided by Rédei & Tsai (2012).

Diversity and distribution. The genus is distributed in the East Palaearctic and neighbouring regions of Indomalaya (southern China, Indochina, Ryukyu Archipelago, Taiwan). Nineteen species, including the new species described below, are currently recognized (see Discussion), although due to taxonomic problems this number will probably be reduced.

Haematoloecha merkli Santillan & Rédei, new species

Figs. 1–22

Type material. Holotype: ♂, “VIETNAM, Yen Bai Prov., Mu \ Cang Chai Distr., Che Tao \ commune, Mu Cang Cai \ Species & Habitats Cons. Area, \ 21.7641°N, 104.0430°E,” [printed], “around Cong Troi (Gate to \ Heaven) Pass, 2040 m, upper \ montane evergreen forest, \ swept & hand-collected, \ 24–29.IX.2016 (#11), \ Ottó Merkl & Phu Pham Van” [printed]; intact, mounted on card, deposited in the Hungarian Natural History Museum, Budapest.

Paratypes (15 ♂♂ 2 ♀♀): with same locality labels as holotype (5 ♂♂ in the Hungarian Natural History Museum, 2 ♂♂ in the National Museum of Natural Science, Taichung); “VIETNAM, Yen Bai Prov., Mu \ Cang Chai Distr., Che Tao \ commune, Mu Cang Cai \ Species & Habitats Cons. Area, \ 21.7695°N, 104.0435°E,” [printed], “around Cong Troi (Gate to \ Heaven) Pass, 2000 m, upper \ montane evergreen forest, \ hand-collected in forest at \ night, 24–29.IX.2016 (#12), \ Ottó Merkl & Phu Pham Van” [printed] (8 ♂♂ 2 ♀♀ in the Hungarian Natural History Museum). All type specimens were provided with adequate type labels.

Diagnosis. This species is easily differentiated from all described species of *Haematoloecha* due to its microptery (Figs. 1–6), the distinct papillar elevations on the anterior lobe of the pronotum that contrast in colour with the surrounding areas (Figs. 7, 9), and the conspicuously bicoloured legs (dark-brown, with femoro-tibial articulations broadly yellow to red) (Figs. 11, 12). A detailed analysis of the diagnostic characters of the species is provided under Discussion.

Description. Micropterous male and female.

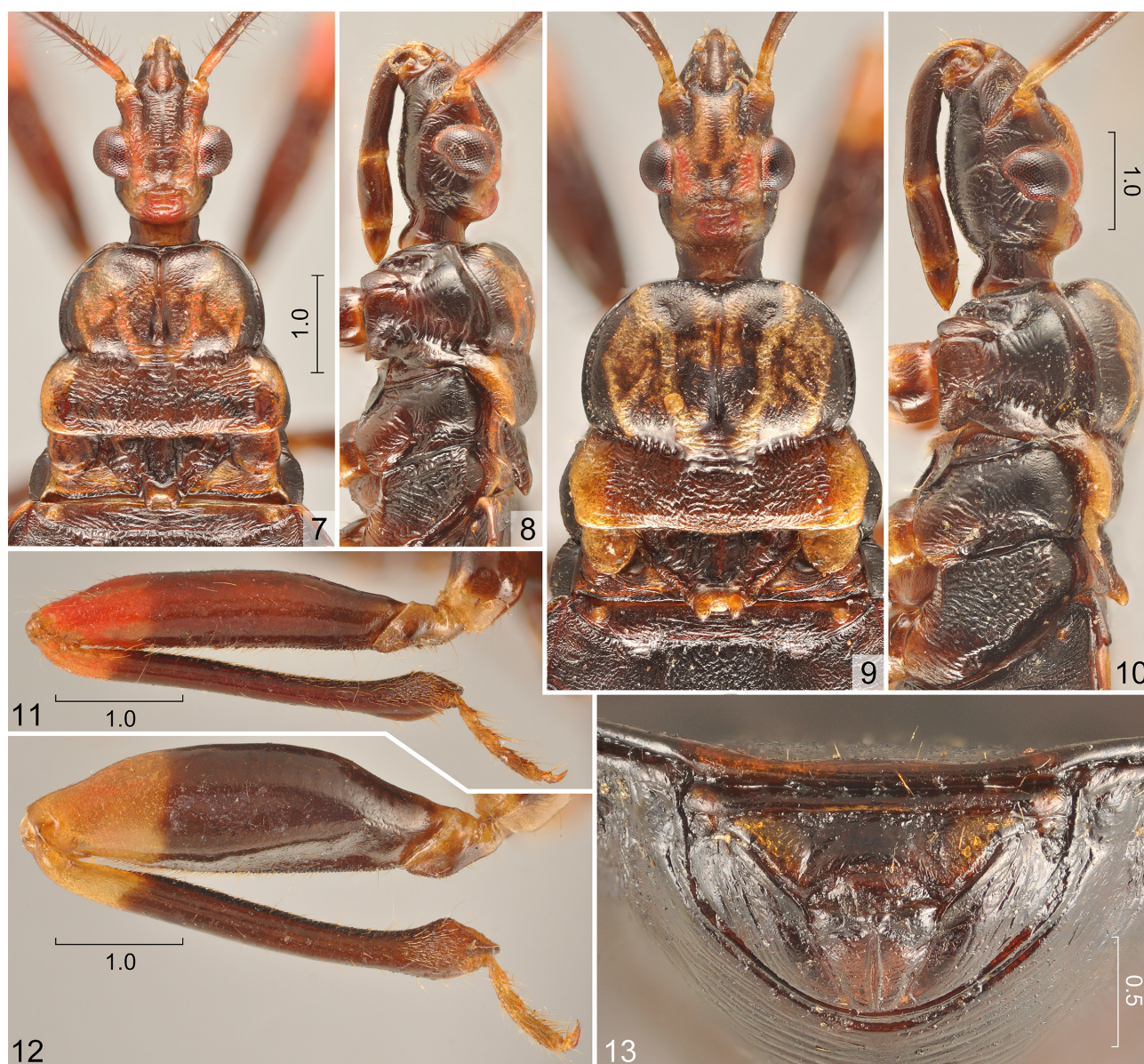
Colour. General colour brown; dorsal surface of head reddish-brown, median third of anterior lobe (between submedian longitudinal sulci, particularly between antenniferous tubercles) clypeus, and labrum distinctly darkened; posterior lobe pale-brown, with distinct reddish pigmentation on ocellar tubercle; ventral surface (ventrad of level of antenniferous tubercles, including entire maxillary plates and bucculae) dark-brown to blackish-brown; labium dark-brown, distal portions of visible labiomeres somewhat lighter; antenna dark-brown, flagellum gradually lighter towards apex, scape with a distinct light-yellowish annulus proximally; pronotum dark-brown, papillar elevations of anterior lobe yellow (frequently with distinct reddish shade), humeral lobes of posterior lobe yellow in their entire length; scutellum dark-brown; mesothoracic wings yellow; exposed portions of metathorax yellowish-brown, distinctly darker-brown posteriorly; thoracic pleuron and sternum dark-brown, sometimes with more or less distinct paler suffusion on supracoxal lobes and on prosternal stridulatory sulcus; legs dark-brown, proximal and distal parts of coxae and proximal parts of trochanters yellowish, femora with more or less distinct dark suffusion proximally and with a contrasting reddish annulus distally (occupying approximately distal third of prothoracic, distal fourth of mesothoracic and somewhat less than distal fifth of metathoracic femora), pro- and mesothoracic tibiae with pale-yellow annulus proximally, metathoracic tibia entirely pale-yellow or light-brown with pale-yellow annulus proximally; tarsi pale-yellow; abdomen dark-brown dorsally and ventrally, tergum I yellowish anteriorly, laterotergites II–VII each yellowish anteriorly, ventral surface also with corresponding yellowish spots at anterior portion of each sternum laterally; tergum IX of female yellowish laterally.

Structure. Body surface and vestiture. Body rather dull dorsally, subshining ventrally. Integument of head more or less deeply transversely rugose dorsally and ventrally, rugae particularly deep and coarse on submedian sulci of head and elevated median area enclosed by them and ventrolaterad to eyes; anterior lobe of pronotum with distinct, broad, conspicuous, symmetrically arranged papillar elevations on disc, transversely rugose on lateral surface neighbouring lateral carina; posterior lobe of pronotum deeply, coarsely rugose between humeral sulci, humeral lobes smooth; scutellum and thoracic pleuron coarsely rugose; abdominal tergum I finely, tergum II strongly (but less deeply than posterior lobe of pronotum), remaining pregenital terga and all exposed pregenital ventrites very finely rugose. Body glabrous dorsally, with scattered, semi-erect whitish setae on clypeus and base of labium, with several isolated, long and fine whitish setae arranged symmetrically as follows: a pair below ventral margin of maxillary plate, two pairs (an anterior and a posterior) close to mesal margin of eye, and three pairs on anterior part

of anterior lobe of pronotum; antenna with short, fine, adpressed to semi-erect, whitish pilosity intermixed with sexually dimorphic longer setation: ♂ with several strongly erect to perpendicular brownish setae several times (on pedicel about 3–4 times) longer than diameter of antennomeres, ♀ with much less strongly erect yellowish setae restricted to pedicel and flagellum (their density increasing towards distal part of antenna), somewhat shorter than in



FIGURES 1–6. *Haematoloecha merkli* sp. nov. 1, male, dorsal view; 2, same, lateral view; 3, same, ventral view; 4, female (individual with posterior part of abdomen slightly distorted), dorsal view; 5, same, lateral view; 6, same, ventral view. Scale bar in mm.

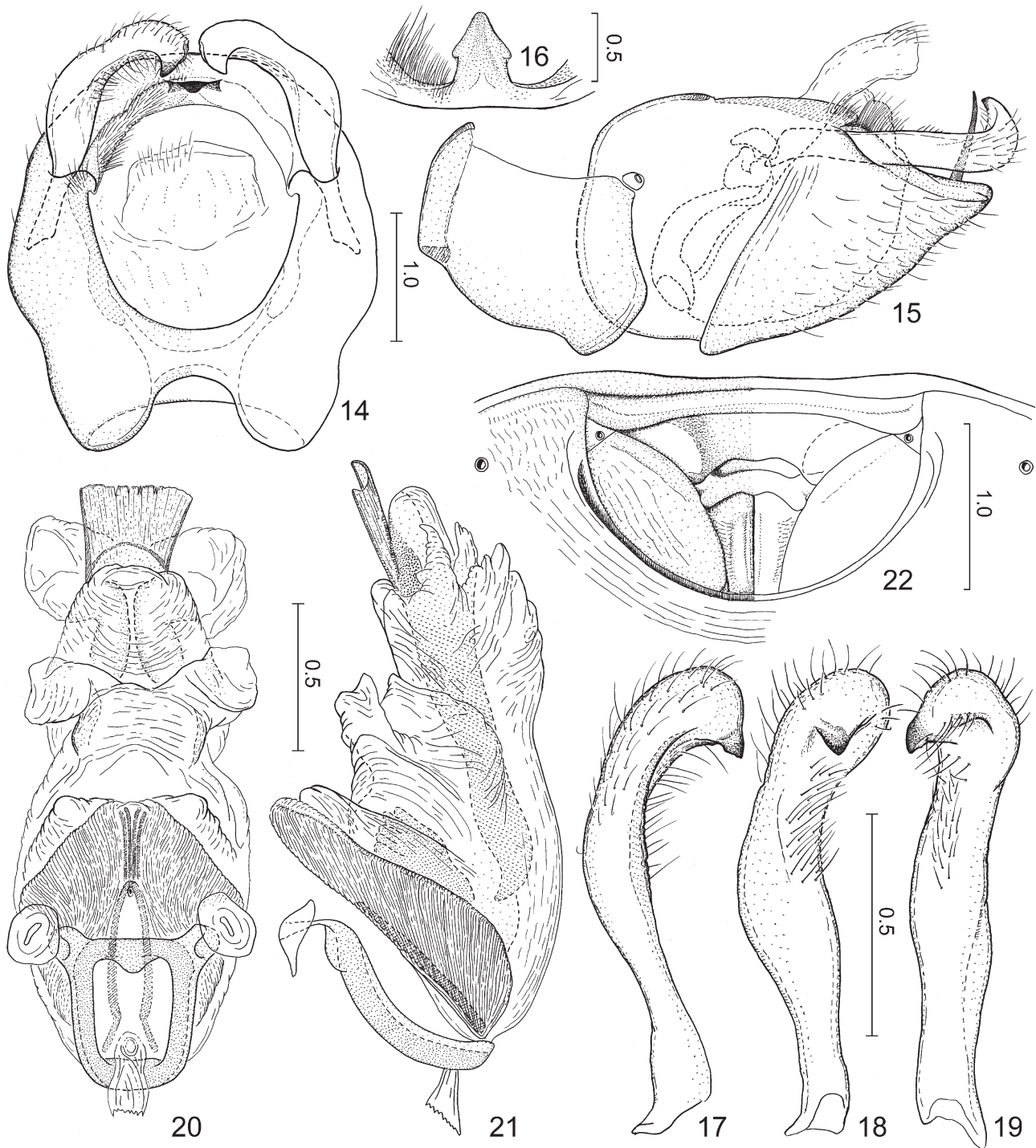


FIGURES 7–13. *Haematoloecha merkli* sp. nov. **7**, head and thorax of male, dorsal view; **8**, same, lateral view; **9**, head and thorax of female, dorsal view; **10**, same, lateral view; **11**, left prothoracic leg of male, posterior (external) view; **12**, left prothoracic leg of female, posterior (external) view; **13**, terminalia of female, posterior view. Scale bars in mm.

male (on pedicel about 2–3 times longer than its diameter); coxae and trochanters with a few relatively long, semi-erect setae, femora and tibiae with short, relatively sparse, fine, semi-erect pilosity (more dense ventrally), with a few isolated long, almost perpendicular setae scattered on femora and tibiae anteroventrally and posteroventrally, most conspicuously on prothoracic legs; abdomen glabrous dorsally, with a few fine, semi-erect setae scattered on its venter.

Head (Figs. 7–10) 2.00–2.05 (♂) / about 1.75 (♀) times as broad as interocular distance, anteocular part 0.5–0.6 (♂) / about 0.5 (♀) times as long as length of head; clypeus relatively low, gradually and continuously declivous anteriad; median portion of head between submedian longitudinal sulci highly elevated, strongly convex, but depressed immediately anteriad of transverse sulcus (more strongly in ♂); eye relatively small, semiglobular, approaching but not reaching dorsal, far remote from ventral outline of head in lateral view, length subequal to length of postocular portion of head in dorsal view in both sexes; ocellar lens reduced to various extent, usually lacking, but remnants might be present (usually only on one side); pedicel 1.25–1.30 (♂) / 1.15–1.20 (♀) times as long as scape; first visible labiomere reaching to about level of anterior margin of eye. *Thorax* (Figs. 7–10).

Pronotum short, 1.25–1.30 (♂) / 1.30–1.45 (♀) times as broad across humeri as total length; anterior lobe 1.40–1.55 (♂) / 2.0–2.3 (♀) times as long as posterior lobe, dorsally convex, elevated emerging only slightly (♂) or highly (♀) above level of posterior lobe; posterior lobe shortened, about 1.15–1.25 (♂, ♀) times as broad across humeri as greatest width of anterior lobe; scutellum short and broad, slightly shorter along midline than posterior lobe of pronotum, apical prongs more pronounced in males, shorter and more obtuse in females; fore wing strongly reduced, forming a uniformly sclerotized, almost circular flap, distance between its tip and posterior margin of metanotum about half its length. *Legs*. Fore femur of considerably variable thickness in both sexes, in males relatively narrow, with dorsal and ventral outlines subparallel in lateral view, 4.30–5.45 times as long as its greatest diameter, whereas that of females strongly thickened, with dorsal outline distinctly convex in lateral view, 3.10–3.45 times as long



FIGURES 14–22. *Haematoloecha merkli* sp. nov. 14, male, genital capsule, dorsal view; 15, male, genital capsule and segment VIII, left lateral view; 16, male, dorsoapical process of genital capsule, posterior view; 17–19, right paramere, three different views; 20, phallus, dorsal view; 21, same, left lateral view; 22, terminalia of female, posterior view. Scale bars in mm.

as its greatest diameter; ventral face anteroposteriorly compressed, forming a distinct longitudinal carina that is coarsely granulate along its entire length; fore tibia straight, relatively gracile, fossula spongiosa occupying about apical 10% of its total length in both sexes (Figs. 11, 12). *Abdomen* elliptical, distinctly broader in females than in males, 1.3–1.4 (♂) / 2.0–2.3 (♀) times as long as its greatest width, males with distinct cleft at posterior margin of abdomen; posterior margin of tergum VII distinctly notched at midline in males.

Male genitalia (Figs. 14–21). Genital capsule (Figs. 14–15) markedly constricted at level of dorsal margin of posterior aperture in dorsal view, dorsoapical process relatively robust in most exposed view, apically blunt, gradually diverging towards base and forming a pair of small tubercles laterally somewhat below middle, then forming a sub-parallel stem (Fig. 16). Paramere (Figs. 17–19) rather robust, terminal part broadly globose, apical denticle large and placed distinctly subapically. Phallus (Figs. 20–21) with dorsal sclerotized plate broad, heavily rugulose; phallosome with a pair of short lobes below dorsal sclerotized plate; evaginated endosoma externally with a pair of short, rounded lobes distally below phallosome, and with a broad, dorsoventrally flattened, elongate inner sclerite characteristic for the genus, moderately broadened distally and truncate apically. *Female genitalia* (Fig. 21). Tergum VIII very short, anterior and posterior margins nearly straight; tergum IX nearly vertical, surface depressed sublaterally on both sides, posterior margin deeply, subtrapezoidally emarginated; valvifer VIII broadly rounded; valvula VIII relatively narrowly exposed, morphological posterior (= topographic dorsal) margin protruding posteriad (= dorsad) medially. Due to the scarcity of available females (only two individuals) the internal female genitalia were not examined.

Measurements (holotype / males, n = 5 / females, n = 2). Body length 8.45 / 7.94–9.06 / 2.41–2.53; length of head 1.95 / 1.63–2.17 / 2.41–2.53, length of anteocular part 1.13 / 1.01–1.13 / 1.17–1.27, width across eyes 1.44 / 1.40–1.55 / 1.64–1.68, interocular distance 0.72 / 0.68–0.77 / 0.93–0.95; lengths of visible labiomeres (I) 1.34 / 1.19–1.37 / 1.52–1.57, (II) 0.66 / 0.64–0.79 / 0.78–0.86, (III) [concealed by coxae, not measured] / 0.31–0.39 / 0.39–0.42; length of scape 2.30 / 1.98–2.30 / 2.01–2.08, pedicel 2.51 / 2.49–2.96 / 2.40–2.41, proximal basiflagellite 1.22 / 1.22–1.25 / 1.17–1.22, distal basiflagellite 0.89 / 0.83–0.97 / 0.88–1.01, distiflagellum (4 distiflagellites combined) 1.47 / 1.68–1.83 / 1.74; length of pronotum 1.95 / 1.87–2.04 / 2.40–2.42, of anterior lobe 1.17 / 1.08–1.23 / 1.56–1.72, of posterior lobe 0.78 / 0.69–0.88 / 0.69–0.86, width of anterior lobe 2.16 / 2.02–2.18 / 2.68–2.81, width of posterior lobe 2.51 / 2.33–2.65 / 3.15–3.48; median length of scutellum 0.62 / 0.62–0.82 / 0.78–0.86, width at base 1.38 / 1.27–1.56 / 1.56–1.73; length of fore wing 0.49 / 0.37–0.62 / 0.42–0.51; length of fore femur 2.84 / 2.82–3.27 / 3.37–3.42, greatest width 0.66 / 0.56–0.70 / 0.98–1.07; length of abdomen (in dorsal view from base to apex of tergum VII) 5.89 / 5.74–6.33 / 7.34–7.42, greatest width 4.26 / 4.07–4.84 / 5.62–5.86.

Etymology. Patronym dedicated to the late Dr. Ottó Merkl, former Curator of Coleoptera at the Hungarian Natural History Museum, who collected the type series.

Distribution. Vietnam (Yên Bái Province).

Discussion

Checklist of species of *Haematoloecha*

Because the latest catalogue of *Haematoloecha* (Maldonado Capriles 1990) did not contain some of the taxonomic changes proposed by prior authors, and new species were described and a considerable number of other taxonomic changes have been proposed subsequently, a checklist of *Haematoloecha* species and subspecies currently considered valid, together with their junior synonyms and distribution, is provided below. Species of which the type material has been examined by the second author, and countries where the type locality of the respective species are currently located, are marked with asterisks; remarks are also made on some unresolved taxonomic problems. For bibliographic data of the original descriptions of the species and the proposed synonymies, the works of Maldonado Capriles (1990), Putshkov & Putshkov (1996), and Rédei & Tsai (2012) should be consulted.

Haematoloecha adachii Tachikawa, 1968 (Japan*, Taiwan)

Haematoloecha bicoloripes Distant, 1919* (Vietnam*)—Probably a junior synonym of *H. rubescens*.

Haematoloecha chapana Distant, 1919* (Vietnam*)—Probably should be removed from *Haematoloecha*; it cannot be placed into any of the described genera of Ectrichodiinae.

Haematoloecha curta Rédei & Tsai, 2012* (Taiwan*)
Haematoloecha delibuta (Distant, 1883)* (Korea, Japan*)
Haematoloecha fokiensis Distant, 1903* (China*)
Haematoloecha iguchii (Matsumura, 1913)* (Japan*)—Probably conspecific with *H. nigricollis* (Tomokuni & Cai 2002).
Haematoloecha limbata Miller, 1954* (Korea, China*)
 = *Haematoloecha aberrans* Hsiao, 1973*
H. merkli sp. nov.* (Vietnam*)
Haematoloecha minor Hsiao, 1973* (China*)
Haematoloecha morosa Breddin, 1903* (Vietnam*)
Haematoloecha nigricollis Matsumura, 1913 (Japan*, Korea?)—Probably a junior synonym of *H. delibuta* (Fukui 1927, Putshkov & Putshkov 1996); probably conspecific with *H. iguchii* (Tomokuni & Cai 2002).
Haematoloecha nigrorubra nigrorubra Distant, 1919 (China, Laos*)
 = *Haematoloecha yunnana* Hsiao & Ren, 1981*
Haematoloecha nigrorubra conspicua Rédei & Tsai, 2012* (Taiwan*)
Haematoloecha nigrorufa Stål, 1867*—Korea, Japan*, China, Vietnam, Taiwan
 = *Ectrichodia includens* Walker, 1873
 = *Ectrichotes nigriventris* Fallou, 1887*
 = *Haematoloecha nigrorufa* f. *fusca*: Hsiao & Ren (1981)—Unavailable name.
 = *Haematoloecha nigrorufa* f. *rufa*: Hsiao & Ren (1981)—Unavailable name.
Haematoloecha obscurata Distant, 1919* (Vietnam*)—Probably should be removed from *Haematoloecha*; it cannot be placed into any of the described genera of Ectrichodiinae.
Haematoloecha puetzi (Kerzhner & Günther, 2004)* (China*)
Haematoloecha pusilla Rédei & Tsai, 2012* (Taiwan*)
Haematoloecha rubescens Distant, 1883* (Japan*, Vietnam)
 = *Haematoloecha laeta* Breddin, 1912*
 = *Haematoloecha laeta* var. *ardens* Breddin, 1912*
Haematoloecha rufithorax (Breddin, 1903)* (Japan*)
 = *Haematoloecha longiceps* Miller, 1956*

Comparison of *Haematoloecha merkli* sp. nov. with related taxa

Character states of *Haematoloecha merkli* sp. nov. recognized as of diagnostic value are discussed below.

(1) **Microptery.**—The new species evidently differs from most of the described species of *Haematoloecha* by its microptery. Although the hemelytra might be shortened to different degree in other congeners, namely *H. rufithorax* (Breddin, 1903) and *H. limbata* Miller, 1953, micropterous morphs (entirely lacking a differentiated membrane) are known in only two other species of the genus: *H. puetzi* (China: Sichuan) and *H. curta* (Taiwan). Both of the latter two species were treated by Rédei & Tsai (2012). The new species is immediately distinguished from them based on its colour (ground colour dark-brown with yellow and red markings on the head, thorax and abdomen dorsally; all legs with broad red annuli surrounding femoro-tibial articulations) (Figs. 1–12); micropterous morphs of the other two species are fairly uniformly dark-brown, at most with some indistinct pale markings in *H. curta* (Rédei & Tsai 2012: 4, figs. 1–2). In addition to colour, the female of *H. puetzi* (male unknown) differs from the female of *H. merkli* sp. nov., among others, by the distinctly larger eyes, different shape of the postocular lobe of the head, relatively longer anterior lobe of the pronotum (as compared to the length to the posterior lobe), mesothoracic wings surpassing the posterior margin of the metanotum, the fore femur being much less thickened, and the legs being covered by a relatively long and dense pilosity (Rédei & Tsai 2012: 5, figs. 8–11). Males and females of *H. curta* conspicuously differ from the respective sexes of *H. merkli* sp. nov. among others by the highly different shape of the head (more elongate, anteriorly less tapering in lateral view, with a dorsally elevated proximal portion of the clypeus in both sexes of *H. curta*) and the smooth anterior lobe of the pronotum. A comparison of specimens of these species suggests that the similarity of their general habitus is superficial, greatly caused by the micropterous condition of the wings, and the three species are probably not closely related phylogenetically.

All other species that are currently placed into *Haematoloecha* are known from macropterous individuals only. The eventual occurrence of micropterous morphs in these species, however, cannot be excluded; the hypothesis that individuals recognized here as pertaining to *H. merkli* **sp. nov.** represent merely micropterous morph of an already described species is considered below. As wing polymorphism always more or less heavily affects the skeletal morphology of all thoracic segments, associating conspecific individuals of different wing morphs might be challenging in Reduviidae (Zhang & Weirauch 2010, Rédei *et al.* 2012). However, wing polymorphism normally has no impact of the characters discussed below; therefore, they are offered as potentially of diagnostic value for distinguishing *H. merkli* **sp. nov.** from these species.

(2) **Colour and sculpture of pronotum.**—In *H. merkli* **sp. nov.** the anterior lobe of the pronotum is provided with distinct, broad, conspicuous, symmetrically arranged papillar elevations that are yellowish, contrasting with the dark-brown ground colour; the posterior lobe is distinctly bicolorous, dark-brown with the humeral lobes yellowish (Figs. 7, 9). In macropterous morphs of all species currently placed into *Haematoloecha*, the surface of the pronotum is smooth and lacks elevated portions. Although different colour patterns occur in the genus (anterior and posterior lobes of the same colour, or only median longitudinal and transverse sulci with dark pigmentation; in other species, the colour of anterior and posterior lobes contrastingly differ; in still other species, the disc and humeral lobes of the posterior lobe also contrastingly differ), none of the described species have a bicolorous anterior lobe of the pronotum.

The anterior lobe of the pronotum is smooth in the majority of the genera of Ectrichodiinae occurring in Indomalaya. A distinctly sculptured anterior lobe occurs in a number of ectrichodiine genera, at least in some, or frequently in all species of e.g. *Cimbus* Hahn, 1831, *Tamaonia* China, 1940, *Scadra* Stål, 1859, *Parascadra* Miller, 1953, *Labidocoris* Mayr, 1865, *Bayerus* Distant, 1904, *Libavius* Distant, 1904 and *Caecina* Stål, 1863, but in all of these the condition is fairly different from *H. merkli* **sp. nov.**, and all of these genera are different from *H. merkli* **sp. nov.** at generic level based on various other characters (cf. Cook 1977). A condition comparable to *H. merkli* **sp. nov.** apparently occurs only in *Rhysostethus* Hsiao, 1973. This genus is currently monotypic; its type species, *Rhysostethus glabellus* Hsiao, 1973 (China: Sichuan), is known from a macropterous male holotype and a micropterous female (the latter is the holotype of *Parascadra breuningi* Kerzhner & Günther, 2004, now a junior synonym of *R. glabellus*) (Rédei *et al.* 2012). The micropterous morph of this species is superficially similar to micropterous individuals of *H. merkli* **sp. nov.**, but it can readily be distinguished by its uniformly dark-brown colour except pale bands along anterior margins of dorsal laterotergites of abdominal segments III–VII, the different shape of the pronotum (with a pair of broad tubercles at the two sides of the meson anteriorly), and most importantly the highly modified prothoracic legs considered as diagnostic for *Rhysostethus*: ventral margin of femur concave, forming a sharp keel, abruptly narrowed subapically, distal portion of tibia strongly bent (Rédei *et al.* 2012: 345, figs. 8 and 11).

(3) **Conspicuously bicolorous legs** (dark, femorotibial articulations broadly yellow to red).—Among the species known from macropterous individuals, legs with areas of more or less contrastingly different colour are found only in *H. rubescens* (Japan, Vietnam), *H. bicoloripes* (Vietnam), and *H. pusilla* (Taiwan). The first two species are highly similar to each other, and based on the re-examination of their type materials, the latter might be a junior synonym of the former. The main difference is the colour of the corium and legs. Because *H. bicoloripes* is known from only a single female syntype (deposited in the Natural History Museum, London), and so far no male matching its colour has been seen from the region, they are here tentatively maintained as distinct at the species level. In *H. rubescens*, the ground colour of the femora and the tibiae are dark-brown, the femora possess a proximal pale-reddish annulus, and the femoro-tibial articulation is broadly bright-red (Rédei & Tsai 2012: 21, figs. 61–63), differing from the condition seen in *H. merkli* **sp. nov.**, where the femora lack a proximal annulus (Figs. 1–6, 11–12). In *H. bicoloripes*, the legs are coloured similarly to *H. rubescens*, but the extreme tips of the femora are dark-brown and the proximal red annulus of the tibiae lacking. In both of these species, the head and the pronotum are uniformly yellowish-red to bright-red, and the pregenital abdominal segments III–VII are decorated with broad black fasciae occupying posterior halves of the respective ventrites. Because no case in which wing polymorphism affects colour of the head, pronotum, abdomen or appendages has been documented in Reduviidae, it can safely be concluded that *H. merkli* **sp. nov.** does not represent the micropterous morph of *H. rubescens* or *H. bicoloripes* but differ from the latter two at species level. In *H. pusilla*, the legs are provided with a broad red annulus on the femoro-tibial articulations (Rédei & Tsai 2012: 4, fig. 3) in a way that is fairly similar to the condition seen in *H. merkli* **sp. nov.**; the small size (8–10 mm, vs. in 9.5–12.5 mm in *H. merkli* **sp. nov.**) and the relatively elongate head

(Rédei & Tsai 2012: 10, figs. 25–26, cf. Figs. 7 and 8 for *H. merkli* **sp. nov.**) are also similar in these two species. A comparison of the male genitalia of *H. pusilla*, figured by Rédei & Tsai (2012: 10, figs. 28–30, and 11, figs. 33–38), and *H. merkli* **sp. nov.** (Figs. 14–21) leaves no doubt that the two species are different at the species level: the shape of the genital capsule (lateral outline more strongly constricted close to the anterior aperture in *H. merkli* **sp. nov.**), particularly its dorsoapical process (forming a pair of distinct tubercles laterally somewhat below middle in *H. merkli* **sp. nov.**, lacking in *H. pusilla*), and the paramere (relatively gracile, apical denticle placed near tip in *H. pusilla*, vs. more robust, with apical denticle placed distinctly subapically in *H. merkli* **sp. nov.**) are all markedly different between these species.

Based on a consideration of all species of *Haematoloecha* known from macropterous individuals only, it might be concluded that *H. merkli* **sp. nov.** can safely be recognized as different at species level from all described congeners.

Generic placement and affinities of *Haematoloecha merkli* **sp. nov.**

Although the sculptured anterior lobe of pronotum (Figs. 8, 10) is unique within the genus, *Haematoloecha merkli* **sp. nov.** matches all important characters used for defining *Haematoloecha*, including the shape of head, the lack of a lamelliform process surrounding antennal insertion, the basiflagellum being subdivided to two and distiflagellum to four secondary flagellites, the relative length of labiomeres (apparent labiomere I distinctly longer than II), the median longitudinal sulcus of the pronotum being deep, reaching the collar anteriorly, and closely approaching a series of transverse median ridges on posterior lobe that represent the highly reduced median sulcus of the posterior lobe (due to a general reduction of the posterior lobe, as a result of microptery). Species with a sculptured and smooth anterior lobe of the pronotum occur in some other reduviid genera as well, e.g. *Rhynocoris incertis* (Distant, 1903) has a distinctly sculptured pronotum, whereas in most congeners it is smooth.

Within *Haematoloecha*, the phylogenetic affinities of *H. merkli* **sp. nov.** are not particularly clear. The new species appears to be most similar to *H. pusilla*, a species endemic to Taiwan; this similarity includes a moderate size, a dull-brownish to brownish ground colour of body (instead of bright-red colours that are found in the majority of the other congeners), reddish annuli surrounding the femoro-tibial articulations, a generally similar shape of the head, and a somewhat similar shape of dorsoapical process of the genital capsule. The polarities of these character states (whether they represent apomorphic or plesiomorphic conditions) are unknown, and none of them are considered as particularly convincing to support a phylogenetic relationship between these two species.

The morphology of the new species is consistent with the definition of *Haematoloecha* (Rédei & Tsai 2012), except for its unusual pronotum. Moreover, it exhibits a distinct morphological similarity to *H. pusilla* (a species more clearly pertaining to this genus, lacking the unusual structure of the pronotum). Currently, there seems no reason to treat *H. merkli* **sp. nov.** as different from species of *Haematoloecha* at the generic level. The genus in the current sense, however, seems heterogeneous, including species with distinct specializations of the prothoracic femora (*H. chapana*, *H. obscurata*, and some undescribed species from Indochina); therefore, a future study might conclude that it should be divided into several genera.

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